Full manuscripts and shared theory: a first Topic E audit

Human Compression project

11 October 2026

1 What is compared

The intake is Leemans and Schulte's *Groups of type $L_2(q)$ acting on polytopes* (L083) and *Polytopes with groups of type* $\mathrm{PGL}_2(q)$ (L079). Their declared main results classify the regular polytopes of rank at least four with these groups. Our replacement is the complete common theory U_E and its two complete applications A_1, A_2, together with the required program for the ten universal presentations.

The exact source PDFs and all earlier proof admissions are bound in the accompanying JSON records. The L083 contract includes the rank restriction, the unique marked 11-cell and 57-cell, the universal table and their stated geometry. L079's Theorem 10 gives the unique marked 4-simplex. Its seventeen-case semilinear census and subsequent conjecture are ancillary to that expressly stated main theorem and are not recovered here. Other background and chiral discussions are likewise outside the contracts. Their text is nevertheless counted in the full-source lengths. This comparison concerns recovery of the main results, not every assertion.

2 Lengths and complete acquisition cost

The primary ruler is the number of NFC-normalized, non-whitespace characters extracted by pypdf 6.13.0. We include every source page, including titles, introductions, ancillary material and references, removing only explicitly identified repeated page furniture. All of U_E, A_1, A_2 is charged, without removing the shared theory from the total. The program is counted through its complete rendered appendix, in the same units.

Document or artifact	Characters
Full source L083	20,885
Full source L079	20,908
Complete common theory U_E	18,264
Application A_1	8,213
Application A_2	5,737
Required shared program	4,754

Thus the full-manuscript written comparison is

$$\frac{20\,885 + 20\,908}{18\,264 + 8\,213 + 5\,737} = \frac{41\,793}{32\,214} = 1.297355\ldots.$$

Charging our required program gives the known-cost comparison

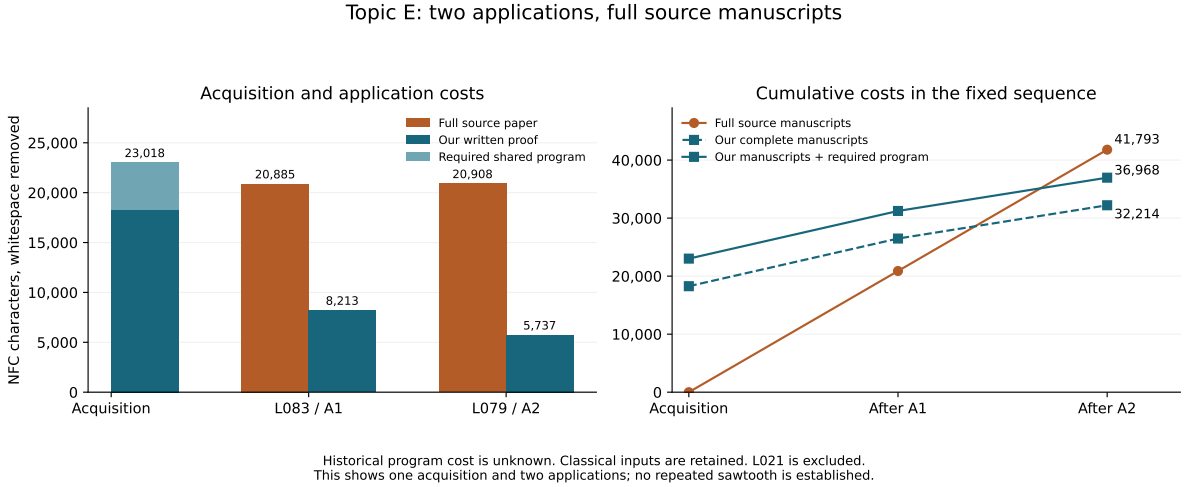
$$\frac{41\,793}{32\,214 + 4\,754} = \frac{41\,793}{36\,968} = 1.130518\ldots.$$

The historical program cost is unknown, so the latter is not an exact all-artifact ratio. Classical external inputs remain explicitly retained; no unconditional library cost or minimum proof length is measured. Conditional on acquiring the complete common theory and shared program, the local factors are $20\,885/8\,213 = 2.5429\ldots$ and $20\,908/5\,737 = 3.6444\ldots$.

3 Sensitivity and the cumulative comparison

The older factor 0.835226 . . . used selected principal-proof passages as the source numerator: 26 906/32 214. It answers a narrower question and remains visible. In particular, the present full-manuscript saving does not establish that our proofs are shorter than those selected source proofs. Across all five fixed text metrics and both NFC and NFKC normalizations, the full-manuscript written factors range from 1.297 to 1.552; the corresponding comparisons including our required program range from 1.131 to 1.403. These are extraction sensitivities, not confidence intervals.

The source PDFs contain 14 and 9 pages; our papers contain 7, 3 and 2 pages, plus a three-page program appendix. Different typography makes page counts supplementary observations rather than the primary ruler. All six PDFs have been checked for nonprinting control characters, replacement characters and private-use characters in their extracted text.



The full acquisition is 18 264 + 4 754 = 23 018. After the two applications, our cumulative known costs are 31 231 and 36 968, compared with source lengths 20 885 and 41 793. The second application repays the measured acquisition under this full-manuscript comparison. There is one acquisition and two applications; a repeated sawtooth pattern has not been established.

4 Audit scope and remaining work

The author measurement and two independent recounts agree. Root's recount recognizes each page header or footer directly, without using the author's extraction spans. All ten metric coordinates and all six document text hashes agree; the complete required GAP program occurs in the charged appendix. Source PDFs and extracted primary texts are not redistributed. The protocol, code, measurements and reviews are published alongside this note.

This audit retains the previously reviewed mathematical contracts. It does not add source recoveries or open-question answers. The later L021 application is excluded: its complete costs and enlarged dependency graphs still require review. The existing theorem graphs identify uses of results, but a complete partition into individually weighted theorem units remains unfinished. The next measurement must include that application's new group-theoretic interface and required computation, as well as all retained proof dependencies.